

A Semi-Supervised Machine Learning Framework for Detecting Political Hoaxes Based on Keyword-Guided Clustering

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Abstract— The spread of political hoaxes poses a significant threat to democratic integrity and social stability. This study proposes an efficient and structured machine learning approach to detect political hoax content. The text data undergoes preprocessing, including normalization, removal of punctuation and numbers, tokenization, and stopword elimination. Numerical representation is generated using Term Frequency–Inverse Document Frequency (TF-IDF), followed by clustering using the K-Means algorithm to group the text based on feature similarity. The core contribution of this research is the development of an automatic labeling strategy based on political keywords, applied after the clustering stage. Unlike previous studies that rely on pre-labeled datasets or stop at unsupervised clustering, this approach introduces a contextual labeling process to distinguish between political and non-political hoaxes by identifying keyword occurrences within the clustered data. The automatically labeled data is then classified using three classical machine learning algorithms: Logistic Regression, Support Vector Machine (SVM), and Naive Bayes. The models are evaluated to assess their ability to accurately detect political hoaxes. Among them, SVM demonstrates the highest and most consistent classification performance, indicating its suitability for this task. The proposed integrated pipeline proves effective in addressing the challenge of unlabeled data. This study contributes methodologically to the development of adaptive and efficient hoax detection systems that are applicable across various socio-political contexts.

Keywords— *political hoax detection, automatic labeling, machine learning.*

I. INTRODUCTION

The development of information and communication technology has changed the paradigm of information dissemination significantly, allowing information to be spread quickly and widely. This is confirmed by [1], in their research which revealed that information and communication technology has a major impact in expanding the reach of service access and increasing efficiency in various sectors.

In relation to information and communication technology, social media has a role in the dissemination of information and distribution of news, as has been proven by [2] that social media as one of the main mediums in news distribution, allows information to spread instantly and widely. However, this change not only brings positive impacts, but also poses major

challenges, one of which is the increasing spread of fake news or hoaxes. Research results [3] reveals the negative impacts of this phenomenon, including the increasing spread of fake news or hoaxes, especially in the political realm that is spread during the election campaign period or before that period. Political hoaxes are often used to manipulate public opinion, influence voter preferences, and damage the reputation of certain individuals or groups. As shown [4] in his research investigating the impact of spreading hoaxes in election campaigns by candidates on voter behavior and election results, especially in competitive elections, the use of hoaxes can affect voter support. This will damage the democratic process in a country that implements democracy, such as the statement [5] which states that the spread of unverified information has the potential to cause social polarization and damage the integrity of democracy.

The spread of fake news, especially during election periods, poses a significant threat to the integrity of information. [6] in his research emphasized that fake news has the potential to change the way people view reality, affect the reputation of institutions and organizations, and can even threaten national security by influencing perceptions of values and risks. Detecting fake news or hoaxes is very important to maintain public trust and the democratic process. Based on the report "Hoax Landscape 2024 Semester 1" from Indonesian Anti-Defamation Society (MAFINDO), 2,119 hoaxes were identified in the first six months of 2024, with a significant increase in political issues directly related to the 2024 General Election and Regional Elections [7].

Political hoaxes are increasingly dominant, reflecting a massive effort to manipulate public opinion. Studies [8], [9] Therefore, there is a strong need for effective approaches to detect hoax content. One popular method that has been applied in prior studies [10],[11],[12],[13],[14] is machine learning. However, most of these approaches rely on pre-labeled datasets, and in many cases, the labeling process is done manually, without incorporating automated labeling strategies.

To address this gap, this study proposes a novel approach that combines K-Means clustering and political keyword analysis to automatically generate labels before the model training phase. This process enables the grouping of data based on feature similarity, followed by contextual labeling to

distinguish between political and non-political hoaxes without the need for manual intervention during the initial stage.

This study is expected to contribute methodologically to the development of more adaptive political hoax detection systems, and to encourage further exploration of detection techniques based on semi-supervised machine learning and context-aware labeling.

In this paper will be discussed in several sessions: Session II will discuss previous research that has similar topics with this research. Session III will discuss the general framework of hoax detection research and introduce the proposed model. Session 4 will discuss the experiments conducted with the proposed model. Session V will present the research results and followed by a discussion related to the findings and session VI will provide conclusions and suggestions for further research.

II. RELATED WORK

The spread of political hoaxes is becoming an increasingly big problem, especially in the run-up to important political events such as elections, and this spread is increasingly massive during the run-up to election campaigns. Various approaches in the field of machine learning have been applied to detect hoaxes in general and even related to politics, with the aim of increasing accuracy and efficiency in identifying fake news.

Research conducted by [12], examined the use of various machine learning techniques, including Naive Bayes, Logistic Regression, Decision Tree, Random Forest, and SVM, in detecting fake news. The results of this study indicate that the Logistic Regression and Random Forest models provide excellent accuracy in distinguishing fake news from real news, with LR 96%, NB 92%, DT 90% RF 94% and SVM 93% accuracy, respectively. This study highlights that simple yet effective text-based techniques, such as Logistic Regression, can provide excellent results in detecting fake news related to political issues.

[15]Rahman et al. (2022) in their study explored a machine learning-based framework for detecting political fake news on social media platforms. Their study compared traditional machine learning models with deep learning using LSTM and Bi-LSTM. The results showed that Bi-LSTM outperformed the other models with 99% accuracy, providing a robust approach in detecting fake news, especially those spread through social media. This study emphasizes the importance of a hybrid approach to address the growing challenge of misinformation across platforms.

On the other hand, [16] propose a machine learning-based framework for detecting fake political speech. They combine linguistic and metadata analysis, and use an SVM model that shows 74% accuracy on the Wild dataset. This study shows that metadata, in addition to linguistic analysis, plays a significant role in detecting political hoaxes, and shows how context is also crucial in identifying fake political speech.

In 2023 the results of [17], have applied the BERT model to detect misinformation in political news. By achieving an accuracy of 79.88%, the BERT model is able to capture context better than traditional methods. This study highlights that understanding semantics and context is crucial in detecting political hoaxes, and BERT is an effective tool in capturing more complex language patterns.

Proposal by [13] create a machine learning-based fake news detection system using a hybrid classifier that combines

Random Forest, XGBoost, and Logistic Regression. This system shows how combining multiple models can improve the accuracy of detecting political hoaxes more effectively. This study underlines the importance of using hybrid models in facing the increasingly complex challenges of detecting hoaxes.

In the study [18], discusses the use of deepfakes in politics, which can be used to manipulate public opinion. This study shows that deepfake technology can change the way fake news is spread, especially in politics, and highlights the major challenges in detecting deepfakes that can mislead the public in choosing candidates or understanding political issues.

Lastly in the context of Indonesian politics, [19] investigates the role of hoaxes in Indonesian politics, especially in the context of the 2017 Jakarta gubernatorial Election. This study uses a memetic practice approach to analyze how hoaxes are spread through social media and influence voters. This study provides important insights into how hoaxes function as a propaganda tool in politics and exacerbate social polarization.

Overall, existing studies show that machine learning techniques, both traditional and deep learning-based, are effective in detecting political hoaxes. Models such as Logistic Regression, SVM, Naive Bayes, and Random Forest have proven to be very good at detecting fake news. Integration of linguistic features, metadata, and context analysis is very important in improving the accuracy of detecting political hoaxes. This study will use a model from an algorithm that has been used by previous researchers in detecting hoaxes in general, but this time it will be associated with the political context in Indonesia so that the dataset used will be clustered, using 2 clusters, namely hoaxes and non-hoaxes, then the hoax cluster is labeled based on political keywords or keywords in Indonesia, hoax clusters that have a tendency with Indonesian political keywords will be labeled as political hoaxes automatically, then modeling will be carried out using the selected machine learning algorithm, finally the model performance evaluation is carried out by measuring metrics such as accuracy, precision, recall, and F1-score.

III. PROPOSED SYSTEM

In this section, we discuss the general framework for hoax detection, covering several main phases, namely dataset collection, feature extraction, feature processing, feature selection, and the application of machine learning algorithms involved in this process. These stages aim to identify patterns in text, distinguish between real and fake news, and improve the accuracy of the model in detecting hoaxes.

Figure 1 presents the proposed system architecture for political hoax detection. The process begins with collecting a dataset of hoax claims from MAFINDO and a list of political keywords. The data undergoes preprocessing steps including cleaning, tokenization, and stopword removal. Feature extraction is then performed using TF-IDF vectorization. The preprocessed vectors are clustered using the K-Means algorithm, and the resulting clusters are labeled based on the presence of political keywords. These labeled samples are then used to train supervised machine learning models. The performance of the models is evaluated using standard

classification metrics, and the final results are analyzed accordingly.

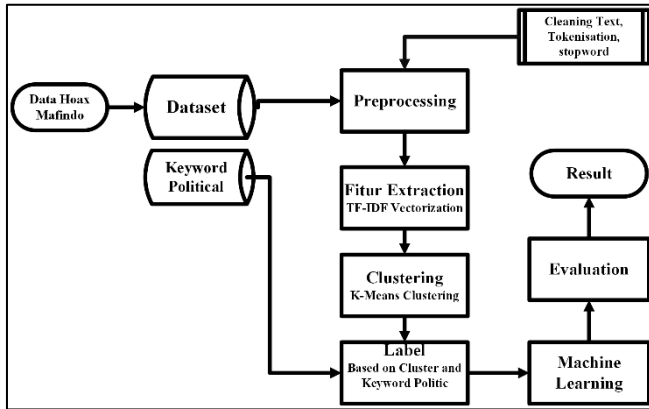


Figure 1 Proposed framework for political hoax detection using keyword-guided clustering and supervised machine learning

IV. METHODOLOGY

In an effort to detect political hoax content in cyberspace, this study proposes a framework based on a machine learning model. Figure 1 shows the main components of the proposed political hoax content detection model. This framework is designed to optimize content detection using various machine learning algorithms and techniques, so that it is expected to be able to increase detection accuracy and reduce error rates.

A. Dataset

The dataset developed in this study was collected from MAFINDO, covering various categories of political hoaxes from 2022 to 2024. The dataset used in this study comes from a collection of news related to political hoaxes in Indonesia. This data is imported from a CSV file and contains information in the form of news text that needs to be further processed for analysis. The dataset used contains unlabeled news, with various topics that can include political and non-political hoaxes. Feature cleaning and extraction are carried out to make it easier for the model to recognize important patterns that can distinguish between political hoax news and non-political hoax news.

B. Preprocessing

At this stage the existing data is prepared so that it can be used in the further analysis process. This process includes several important steps:

- **Lowercasing:** Text is converted to lowercase to avoid duplication of the same word with different capital letters.
- **Removal of Numbers and Punctuation:** Numbers and punctuation are removed because they do not provide relevant information in the text analysis.
- **Tokenization:** Text is broken down into smaller words or tokens using NLTK's word tokenize, which allows the text to be separated into more easily analyzed elements.
- **Stopwords Removal:** Stopwords in Indonesian are removed using the stopwords list from NLTK, to reduce noise and improve analysis accuracy.

The result of this stage is cleaner and more structured text, ready to be used in feature extraction.

C. Feature Extraction

After the text is processed, the next step is to convert the text into a numeric representation that can be used by machine learning algorithms. The method used for feature extraction is TF-IDF (Term Frequency-Inverse Document Frequency). TF-IDF measures the importance of each word in a document relative to the entire dataset. Using TF-IDF Vectorizer from scikit-learn, we limit the number of features to the top 1000 words to improve efficiency and reduce the dimensionality of the very large data.

The result of this stage is a sparse matrix, which represents each document (news) in vector form based on the frequency of existing words, which is then used for the clustering and classification process.

D. Clustering

To identify news groups that have similarities, K-Means clustering is applied. K-Means is an unsupervised learning method used to group data into a number of clusters based on feature similarities [20]. In this study, it is assumed that there are two main clusters:

- Cluster 1: Hoax news
- Cluster 2: Non-hoax news

K-Means groups news based on the similarity of words that often appear in each article. The clustering results show the division of news into two main groups which are then analyzed further.

E. Labeling Based on Clusters and Keywords Politics

After clustering is completed, each cluster is labeled based on the presence and dominance of political-related terms within the news content, particularly the headlines and summary texts. Cluster 1, which frequently includes political keywords such as "election", "party", "president", "candidate", and "campaign", is labeled as "Political Hoax". Cluster 2, which lacks such political context, is labeled as "Non-Political Hoax" or "Valid News", depending on the reliability and framing of the content.

To reduce mislabeling due to ambiguous or incidental political terms, we applied a keyword scoring mechanism. Each news content is assigned a score based on the frequency and TF-IDF weight of predefined political terms. Only when the score surpasses a set threshold is the content considered politically related.

Additionally, contextual disambiguation is used by analyzing the co-occurrence of political keywords within a ± 3 -word window. This ensures terms like "president" are interpreted correctly distinguishing between political references ("president election") and non-political contexts ("president of the club").

This labeling method helps mitigate false positives and enhances the robustness of training data used in the supervised learning stage.

F. Modeling

After the data is labeled, a machine learning model is applied to classify news based on the labels that have been determined. Some of the models used in this study are Logistic Regression, SVM, Naive Bayes

Logistic Regression is a statistical method used to predict the probability that an entity falls into one of two category classes [21]. SVM is a classification algorithm that finds the

optimal hyperplane to separate data into two classes. [21],[22]. Naive Bayes is a probabilistic classification algorithm that is based on Bayes' Theorem and assumes that all input features are independent [21],[22]. In the context of political hoax detection, Naive Bayes is used to classify whether a news item is a political hoax (1) or not a political hoax (0) based on the probability calculated from features such as certain political words in the text. The modeling stages are illustrated in the following script

Pseudocode Model and evaluate

```
1. # Preprocessing: TF-IDF vectorization on cleaned text
2. X = vectorizer.fit_transform(data["clean_berita"])
3. y = data["label"]
4. # Splitting the data into training and testing sets
5. X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
6. # Display class distribution in the test set (important for interpreting metrics)
7. class_counts = y_test.value_counts()
8. total_test = len(y_test)
9. for label, count in class_counts.items():
10.     percent = (count / total_test) * 100
11.     print(f"Label: {label} → Jumlah: {count} ({percent:.2f}%)")
12. # Initialize the models used
13. models = {
14.     "Logistic Regression": LogisticRegression(random_state=42),
15.     "SVM": SVC(random_state=42),
16.     "Naive Bayes": MultinomialNB()
17. }
18. # Train and evaluate each model
19. for model_name, model in models.items():
20.     print(f"Evaluasi Model: {model_name}")
21.     model.fit(X_train, y_train)
22.     y_pred = model.predict(X_test)
23.     print(classification_report(y_test, y_pred))
```

The python script used in this study written above splits the dataset into 70% training data and 30% test data, also using `train_test_split()`, with `random_state=42` to ensure consistent data splitting every time the code is run. Three classification models (Logistic Regression, SVM, and Naive Bayes) are initialized in the dictionary `models`.

After that, for each model, an evaluation is carried out using Stratified K-Fold Cross Validation (5-fold). This technique ensures that the class distribution in each fold remains balanced. The cross-validation results (accuracy scores) are calculated and displayed along with the average accuracy and standard deviation of each model.

Next, each model is trained on the entire training data and tested using the test data. The classification results for each model are evaluated with the Classification Report which includes metrics such as precision, recall, and F1-score. A Confusion Matrix is also displayed to provide a clearer picture of the number of correct and incorrect predictions on the test data.

This evaluation provides a performance comparison of the three models for the classification task and helps to select the most optimal model based on relevant criteria.

V. RESULT AND DISCUSSION

The spread of false information in the digital era is increasingly becoming a global concern, prompting the development of machine learning-based systems to detect fake news. These systems analyze and classify news articles, social media posts, and other content based on their credibility using artificial intelligence and machine learning techniques. This research focuses on political hoax detection, which is a critical issue in today's information ecosystem.

This study uses a publicly available dataset from MAFINDO. We clustered the data using k-means to cluster hoax news or not, then we divided the hoax cluster into two categories and labeled it: Political Hoax (1) and Non-Political Hoax (0). This dataset contains 1406 data with a total of 422 data clustered as hoaxes, then we labeled the data based on political keywords and found 381 data labeled Political Hoax and 41 data labeled Non-Political Hoax. For illustrative purposes, the following are example headlines from each class: (1) Political Hoax: "Gedung Bawaslu Hingga Mobil Plat Merah Dibakar saat Demo Diskualifikasi Prabowo-Gibran" and (2) Non-Political Hoax: "Video Situasi terkini di sekitaran gedung Mahkamah Konstitusi". These examples highlight how politically charged misinformation is phrased differently than neutral or unrelated hoaxes.

The labeled dataset was split into training and test sets using a 70:30 ratio. In the test set, there were 422 news samples in total, consisting of 381 (90.28%) labeled as Political Hoax and 41 (9.72%) as Non-Political Hoax. This class distribution is crucial for interpreting the performance metrics such as precision, recall, and F1-score reported in Table I. Due to the class imbalance, models that perform well on the majority class (Political Hoax) may still struggle with detecting the minority class (Non-Political Hoax), which is reflected in the detailed classification reports presented below the table.

After preprocessing and transforming the data using TF-IDF, we trained three machine learning models: Logistic Regression, Support Vector Machine (SVM), and Naive Bayes, to classify hoax content based on political relevance.

The dataset was split into training and test sets using a 70:30 ratio. In the test set, 90.28% of the samples were labeled as "Hoax Politik" and 9.72% as "Hoax Non-Politik", presenting a clear class imbalance.

Given this distribution, we evaluated the models not only based on accuracy but also precision, recall, and F1-score to ensure fair performance measurement across classes. Among the three, the SVM model demonstrated the strongest performance, correctly classifying all political hoaxes and most non-political hoaxes, as visualized in the confusion matrix (Figure 3). This highlights the robustness of SVM under imbalanced conditions and confirms its suitability for detecting political hoaxes with high reliability.

TABLE 1. Comparison of Results

Model	Precision	Recall	F1-Score	Accuracy
Logistic Regression	0.93	1.00	0.96	0.93
SVM	0.99	1.00	0.99	0.99
Naive Bayes	0.90	1.00	0.95	0.90

In this study, three classification models, namely Logistic Regression, SVM, and Naive Bayes, were evaluated to detect Political Hoaxes. The Logistic Regression model showed

good performance with a precision of 93%, indicating that 93% of all predicted Political Hoaxes were indeed Political Hoaxes. However, there were around 7% false positives, where Non-Political Hoaxes were classified as Political Hoaxes. The recall of this model reached 100%, meaning that all Political Hoaxes in the test data were detected without any false negatives. The F1-Score of 96% showed a good balance between precision and recall, while the overall accuracy reached 93%.

The SVM model, on the other hand, showed a very high level of accuracy with 99% precision, almost no misclassification, and 100% recall, so that all Political Hoaxes were identified without missing. The F1-Score reached 99%, showing excellent performance in maintaining a balance between precision and recall, and the accuracy reached 99%, making it the model with the highest accuracy in this study.

Meanwhile, Naïve Bayes, although its precision is slightly lower at 90%, still has a recall of 100%, successfully detecting all Political Hoaxes without error. The F1-Score of 95% shows good performance, although slightly lower than Logistic Regression and SVM. The accuracy of this model is 90%, slightly lower than the other two models, but still provides adequate results for detecting Political Hoaxes.

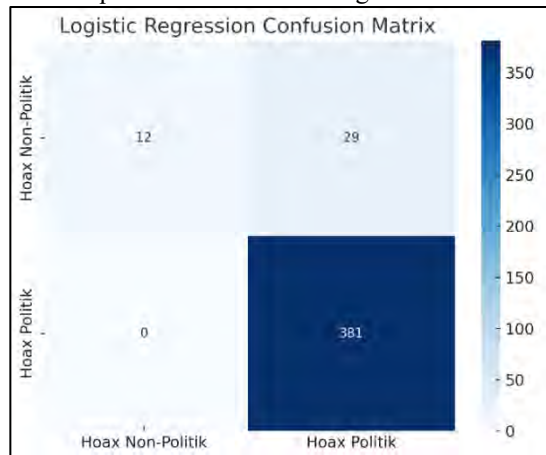


Figure 2 Logistic Regression Confusion Matrix



Figure 3 SVM Confusion Matrix

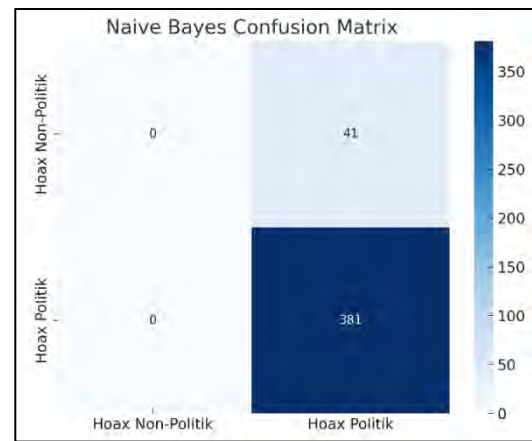


Figure 4 Naive Bayes Confusion Matrix

The confusion matrix for all models (Figures 2, 3, 4) shows a more in-depth performance. As seen in the confusion matrix for SVM, this model shows excellent classification performance with no false negatives or false positives, confirming its effectiveness in detecting Political Hoaxes. Logistic Regression shows slight misclassification of Non-Political Hoaxes, but is very accurate in detecting Political Hoaxes. Naive Bayes, although relatively good in detecting Political Hoaxes, struggles in detecting Non-Political Hoaxes, leading to imbalance in precision and recall.

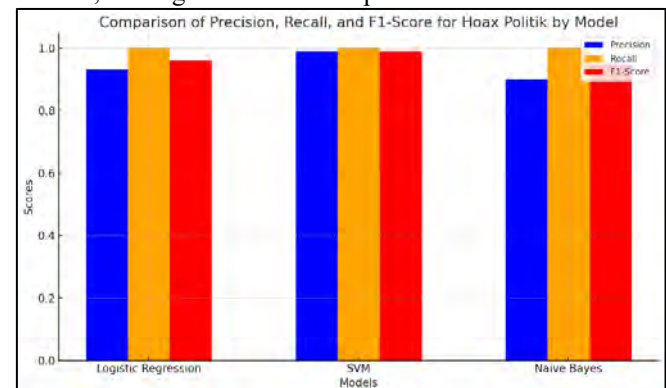


Figure 5 Comparison Precision, Recall, and F1-Score all model

Based on the comparison graph (Figure 5) and the results table (Table 1), it can be concluded that SVM has the highest accuracy in detecting Political Hoaxes, followed by Logistic Regression and Naive Bayes.

The results of this study indicate that SVM is the best performing model for detecting Political Hoaxes, especially in terms of accuracy, precision, and recall. Logistic Regression and Naive Bayes, although effective, show slightly lower precision and recall for Political Hoaxes, especially with Naive Bayes having difficulty detecting Non-Political Hoaxes. To further evaluate model robustness, we plotted the ROC and Precision-Recall curves for each classifier. Figure 6 and Figure 7 show that SVM consistently maintains high AUC values across both metrics, confirming its superiority not only in point metrics but also across thresholds.

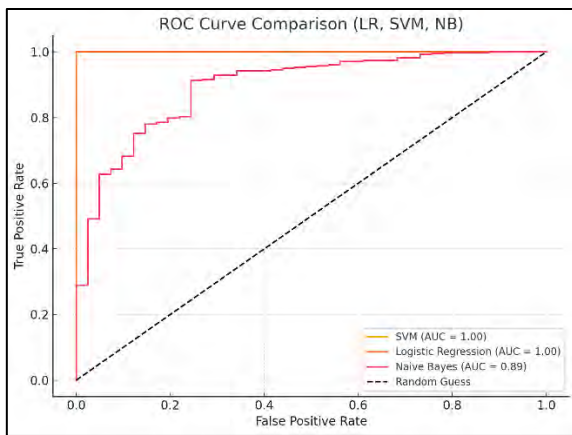


Figure 6 ROC Curve comparison of the classifiers used in detecting Political Hoaxes

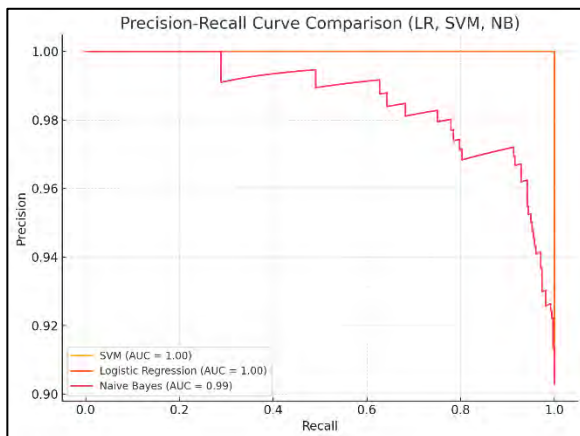


Figure 7 Precision-Recall Curve comparison of SVM, LR, and NB

The results of this study indicate that SVM is the best performing model for detecting Political Hoaxes, especially in terms of accuracy, precision, and recall. Logistic Regression and Naive Bayes, although effective, show slightly lower precision and recall for Political Hoaxes, especially with Naive Bayes having difficulty detecting Non-Political Hoaxes.

Although this study focused on political hoaxes, the proposed hybrid approach combining clustering, keyword-based labeling, and supervised classification has the potential to be generalized to other hoax domains such as health or economic misinformation.

VI. CONCLUSION

This study demonstrates that despite utilizing classical machine learning algorithms such as SVM, Naive Bayes, and Logistic Regression, the structured approach developed combining text preprocessing, TF-IDF-based feature extraction, K-Means clustering, and automatic labeling based on curated political keywords yields promising results in detecting political hoaxes.

The proposed model serves as an efficient yet adaptive framework, allowing classical models to achieve optimal performance under limited data conditions. Unlike deep learning-based approaches that require large datasets and high computational resources, this method provides a lightweight yet accurate solution.

Among the models tested, SVM achieved the highest and most consistent classification performance, indicating that a

systematic pipeline can effectively compensate for limitations in model complexity and data availability.

The main contribution of this study lies in the development of an automatic labeling strategy based on political keywords, integrated within a comprehensive analysis pipeline. This approach not only simplifies the annotation process but also lays the groundwork for the development of broader, more adaptive, and efficient hoax detection systems in the future.

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